

Mass–momentum velocity disparity in compressible fluid flow

Thermodynamic consistency in the slip hypersonic regime

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“The law that entropy always increases—the second law of thermodynamics—holds, I think, the supreme position among the laws of Nature. If someone points out to you that your pet theory of the universe is in disagreement with Maxwell’s equations—then so much the worse for Maxwell’s equations. If it is found to be contradicted by observation—well, these experimentalists do bungle things sometimes. But if your theory is found to be against the second law of thermodynamics I can give you no hope; there is nothing for it but to collapse in deepest humiliation.”

Arthur S. Eddington, *The Nature of the Physical World* (1928)

Compressible fluid flow: Classical description

- Navier–Stokes–Fourier (NSF) equations
- three balance laws
- primary unknown fields

$$\varrho, \quad \vartheta, \quad \boldsymbol{v}$$

- equations of state (pressure, entropy)
- constitutive relations (viscous stress, heat flux)

Single velocity field: $\boldsymbol{v}$

Why question the classical paradigm?

- thermophoresis;
- thermal transpiration;
- diffusiophoresis;
- low-Reynolds-number phoretic motion;
- shock-wave structure;
- rarefied-gas flows.
- microfluidic channels;
- binary diffusion in mixtures;
- tracer versus mass velocity;
- diffuse volume flow across walls;
- diffusion-induced stresses in solids.

Brenner's bivelocity framework

Mass balance

$$\partial_t \varrho + \operatorname{div}(\varrho \mathbf{v}_m) = 0.$$

Momentum balance

$$\partial_t(\varrho \mathbf{v}_v) + \operatorname{div}(\varrho \mathbf{v}_v \otimes \mathbf{v}_m) - \operatorname{div}(\mathbf{S} - p\mathbf{1}) = 0.$$

Energy balance

$$\partial_t(\varrho e) + \operatorname{div}(\varrho e \mathbf{v}_m) + \operatorname{div}(\mathbf{q} - p\mathbf{j}_v) - \operatorname{div}((\mathbf{S} - p\mathbf{1})^\top \mathbf{v}_v) = 0.$$

$$\mathbf{j}_v = \mathbf{v}_v - \mathbf{v}_m,$$

$$\mathbf{S} = \mu(\operatorname{grad} \mathbf{v}_v + (\operatorname{grad} \mathbf{v}_v)^\top), \quad \mathbf{q} = -\kappa \operatorname{grad} \vartheta, \quad \mathbf{j}_v = \iota \operatorname{grad} \ln \varrho.$$

$$\iota \equiv 0 \quad \Longrightarrow \quad \text{Navier–Stokes–Fourier}$$

Corrected bivelocity framework

Mass balance

$$\partial_t \varrho + \operatorname{div}(\varrho \mathbf{v}) = 0.$$

Momentum balance

$$\partial_t(\varrho \mathbf{u}) + \operatorname{div}(\varrho \mathbf{u} \otimes \mathbf{v}) - \operatorname{div}(\mathbf{S} + \tfrac{1}{2} \varrho \mathbf{u} \wedge \mathbf{v} - p \mathbf{1}) = 0.$$

Energy balance

$$\partial_t(\varrho e) + \operatorname{div}(\varrho e \mathbf{v}) + \operatorname{div}(\mathbf{j}_\varepsilon - (\mathbf{S} + \tfrac{1}{2} \varrho \mathbf{u} \wedge \mathbf{v} - p \mathbf{1})^\top \mathbf{u}) + \tfrac{1}{2}(\varrho \mathbf{u} \wedge \mathbf{v}) : \mathbf{W}' = 0.$$

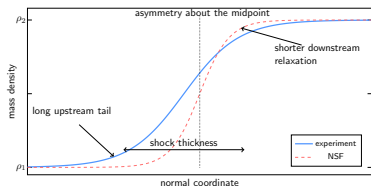
Constitutive relations ($\mathbf{D}' = \operatorname{sym} \operatorname{grad} \mathbf{u}$, $\mathbf{W}' = \operatorname{skw} \operatorname{grad} \mathbf{u}$, $\mathbf{w} = \mathbf{u} - \mathbf{v}$)

$$\mathbf{S} = 2\mu \mathbf{D}'^0 + \zeta(\operatorname{tr} \mathbf{D}') \mathbf{1}, \quad \mathbf{j}_\varepsilon = \mathbf{q} - p \mathbf{w}, \quad \mathbf{q} = -\kappa \operatorname{grad} \vartheta, \quad \mathbf{w} = \iota \operatorname{grad} p.$$

Residual dissipation inequality

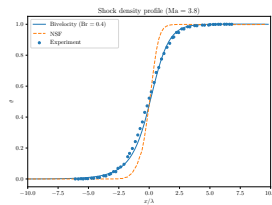
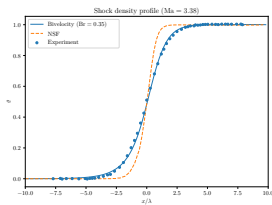
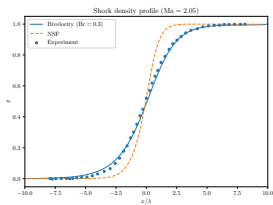
$$2\mu |\mathbf{D}'^0|^2 + \zeta(\operatorname{tr} \mathbf{D}')^2 + \kappa \vartheta |\operatorname{grad} \ln \vartheta|^2 + \iota |\operatorname{grad} p|^2 \geq 0.$$

Steady one-dimensional shock in a monatomic gas



Navier–Stokes–Fourier predictions

- shock thickness too narrow;
- incorrect density asymmetry.

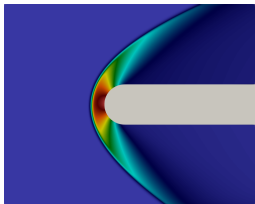


Experimental data: Alsmeyer, *Journal of Fluid Mechanics* (1976)

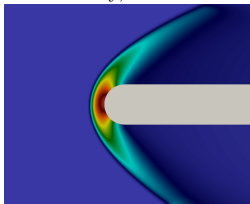
Flow fields around the blunt plate

Pressure

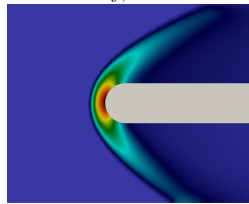
Navier-Stokes-Fourier



Bivelocity, $Br = 0.020$

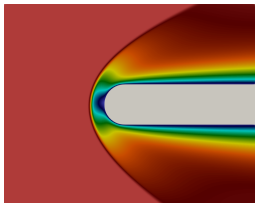


Bivelocity, $Br = 0.050$

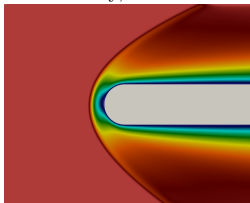


Velocity magnitude

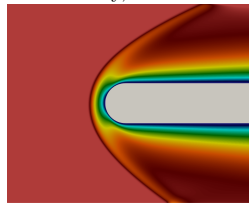
Navier-Stokes-Fourier



Bivelocity, $Br = 0.020$



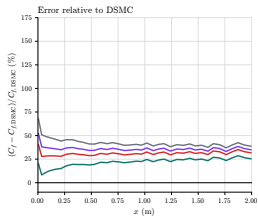
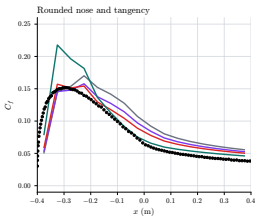
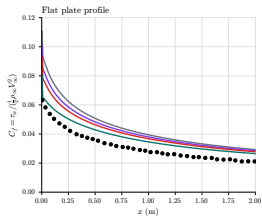
Bivelocity, $Br = 0.050$



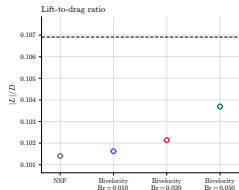
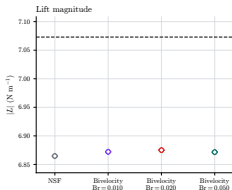
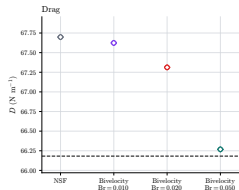
Two-dimensional blunt plate at Mach 15

Upper-surface skin friction: $R_n = 0.4 \text{ m}$, $Ma = 15$, $\alpha = 5^\circ$
Spatial comparison against DSMC from Ou & Chen (2020), Fig. 5(d)

● DSMC (published) — NSF Bivelocity, Br = 0.010 Bivelocity, Br = 0.020 Bivelocity, Br = 0.050



Refined-grid clean-start integral loads against DSMC: $R_n = 0.4 \text{ m}$, $Ma = 15$, $\alpha = 5^\circ$



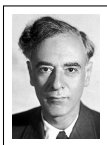
Stats on the blunt plate problem

Refined-grid predictions at $t = 13$; loads averaged over the final 0.05 time units.

Model	Drag D [N m^{-1}]	Lift magnitude $ L $ [N m^{-1}]	Lift-to-drag $ L /D$	C_f RMS error
NSF	67.6983 (+2.29%)	6.8647 (−2.95%)	0.101401 (−5.15%)	42.70%
Bivelocity, $\text{Br} = 0.010$	67.6236 (+2.18%)	6.8720 (−2.84%)	0.101622 (−4.95%)	36.20%
Bivelocity, $\text{Br} = 0.020$	67.3124 (+1.71%)	6.8750 (−2.80%)	0.102135 (−4.47%)	31.07%
Bivelocity, $\text{Br} = 0.050$	66.2679 (+0.13%)	6.8715 (−2.85%)	0.103693 (−3.01%)	22.07%
DSMC	66.1820	7.0730	0.106910	—

Values in parentheses are relative errors with respect to DSMC. The C_f metric is the RMS relative error over the upper flat surface.

From Landau-Teller to mode-wise thermodynamics



Lev Landau



Edward Teller



Chul Park

A progression of ideas

Landau-Teller relaxation $\longrightarrow$ Park's 2T model $\longrightarrow$ Mode-wise thermodynamics

- Landau and Teller: vibrational energy relaxes on its own finite time scale.
- Park: two temperatures made that lag computationally useful in reacting hypersonic flow.
- Here: mode-wise balances are tied together by one dissipation inequality.

From a successful closure to a general thermodynamic framework.

The Fick–Fourier pairing in nonequilibrium thermodynamics

Common Fick–Fourier closure

$$\mathbf{j}_\alpha = -\varrho D_\alpha \text{grad } \varphi_\alpha, \quad \mathbf{q} = -\kappa \text{grad } \vartheta.$$

The fluxes are prescribed independently.

This closure is not, in general, thermodynamically consistent.

Hansen–Fisher (2019)

Corrected Fick–Fourier fluxes conserve mass but need not be entropy stable.

They supplied alternatives with barodiffusion and reciprocal Soret–Dufour coupling for **ideal-gas mixtures** [7].

Here

Weighted modal chemical-potential gradients drive diffusion. A symmetric positive-semidefinite mobility guarantees nonnegative dissipation for **general mixture free energies and arbitrary modal decompositions**.

The second law completes the modal theory

What must be assigned

- modal energies, entropies and temperatures;
- one energy balance for every mode;
- stress power, heat and chemical-potential fluxes;
- reaction, supply and inter-mode exchange terms.

What consistency requires

- exact recovery of the total energy balance;
- cancellation of internal exchanges;
- constitutive restrictions from one entropy imbalance;
- nonnegative residual entropy production.

The second law distinguishes necessary restrictions from constitutive closure.

Temperatures follow the selected energy decomposition

The thermodynamic framework prescribes a set of energy modes $\mathcal{D}$, with

$$\varepsilon = \sum_{i \in \mathcal{D}} \varepsilon_i, \quad \eta = \sum_{i \in \mathcal{D}} \eta_i, \quad \vartheta_i > 0.$$

Two temperatures

$$\mathcal{D}_{2T} = \{\mathbf{tr}, \mathbf{ve}\}, \quad \mathbf{m} = \mathbf{tr}.$$

Four temperatures

$$\mathcal{D}_{4T} = \{\mathbf{t}, \mathbf{r}, \mathbf{v}, \mathbf{e}\}, \quad \mathbf{m} = \mathbf{t}.$$

Modal internal-energy balance

$$\varrho \dot{\varepsilon}_i = \mathbf{T}_i : \mathbf{L} - \operatorname{div} \left(\mathbf{q}_i + \sum_{\alpha \in \mathcal{C}} \mu_{\alpha i} \mathbf{j}_{\alpha} \right) + \sum_{\alpha \in \mathcal{C}} \mu_{\alpha i} \tau_{\alpha}^{\text{ext}} + \pi_i + r_i.$$

$\mathbf{T}_i$ allocates stress power, $\mathbf{q}_i$ is the modal heat flux, and $\mu_{\alpha i} \mathbf{j}_{\alpha}$ transports chemical-potential energy. The exchange π_i transfers energy between modes. The viscous stress belongs to the distinguished mechanical mode $\mathbf{m}$; nondissipative pressure may be distributed among density-dependent modes.

Aggregation and entropy constrain all modes

Energy aggregation

$$\sum_{i \in \mathcal{D}} \pi_i = 0, \quad \mu_\alpha = \sum_{i \in \mathcal{D}} \mu_{\alpha i}.$$

The modal balances recover exactly the total internal-energy balance.

Clausius–Duhem inequality

$$\varrho \sum_{i \in \mathcal{D}} \dot{\eta}_i \geq \sum_{i \in \mathcal{D}} \frac{r_i}{\vartheta_i} - \operatorname{div} \left(\sum_{i \in \mathcal{D}} \frac{\mathbf{q}_i}{\vartheta_i} \right).$$

One entropy inequality constrains all modes together.

For $\psi_i = \hat{\psi}_i(\varrho, \vartheta_i, \{\varphi_\alpha\}_C)$, Coleman–Noll gives

Aggregate restrictions

$$\eta_i = -\partial_{\vartheta_i} \psi_i, \quad \sum_{i \in \mathcal{D}} \frac{\mu_{\alpha i} - \mu_{\sigma i}}{\vartheta_i} = \sum_{i \in \mathcal{D}} \frac{1}{\vartheta_i} \partial_{\varphi_\alpha}^{(\sigma)} \psi_i, \quad \sum_{i \in \mathcal{D}} \frac{p_i}{\vartheta_i} = \varrho^2 \sum_{i \in \mathcal{D}} \frac{1}{\vartheta_i} \partial_\varrho \psi_i.$$

Composition must be treated intrinsically on the simplex

Mass fractions satisfy $\sum_{\alpha \in \mathcal{C}} \varphi_{\alpha} = 1$, so admissible variations obey

$$\sum_{\alpha \in \mathcal{C}} \delta \varphi_{\alpha} = 0.$$

For a reference constituent σ , the Larché–Cahn derivative is

$$\partial_{\varphi_{\alpha}}^{(\sigma)} \psi_i := \partial_{\varphi_{\alpha}} \psi_i - \partial_{\varphi_{\sigma}} \psi_i.$$

The intrinsic mode-wise closure becomes

$$\mu_{\alpha i}^{(\sigma)} = \mu_{\alpha i} - \mu_{\sigma i} = \partial_{\varphi_{\alpha}}^{(\sigma)} \psi_i.$$

Chemical-potential differences are independent of the ambient extension.

Simple closures make every residual mechanism dissipative

Transport

$$\mathbf{T}^{\text{vis}} = 2\mu\mathbf{D}^0 + \zeta(\text{tr}\mathbf{D})\mathbf{1},$$

$$\mathbf{q}_i = -\kappa_i \text{grad } \vartheta_i,$$

$$\mathbf{J}_\alpha = -\sum_{\beta \in \mathcal{C}} M_{\alpha\beta} \sum_{i \in \mathcal{D}} \frac{\text{grad } \mu_{\beta i}}{\vartheta_i}.$$

Reaction and exchange

$$-\sum_{\alpha, i} \frac{\mu_{\alpha i}}{\vartheta_i} \tau_\alpha^{\text{reac}} \geq 0,$$

$$\sum_{i \in \mathcal{D}} \frac{\pi_i}{\vartheta_i} \geq 0.$$

Residual dissipation inequality

$$\begin{aligned} & \frac{2\mu|\mathbf{D}^0|^2 + \zeta(\text{tr}\mathbf{D})^2}{\vartheta_{\text{m}}} + \sum_{i \in \mathcal{D}} \kappa_i |\text{grad } \ln \vartheta_i|^2 \\ & + \sum_{\alpha, \beta \in \mathcal{C}} \left(\sum_{i \in \mathcal{D}} \frac{\text{grad } \mu_{\alpha i}}{\vartheta_i} \right) \cdot M_{\alpha\beta} \left(\sum_{j \in \mathcal{D}} \frac{\text{grad } \mu_{\beta j}}{\vartheta_j} \right) \\ & - \sum_{\alpha \in \mathcal{C}} \sum_{i \in \mathcal{D}} \frac{\mu_{\alpha i}}{\vartheta_i} \tau_\alpha^{\text{reac}} + \sum_{i \in \mathcal{D}} \frac{\pi_i}{\vartheta_i} \geq 0. \end{aligned}$$

A relaxation network conserves energy and produces entropy

For every connected pair of modes $\{i, j\} \in \mathcal{E}$, prescribe

$$Q_{ij} = \Lambda_{ij}(\vartheta_j - \vartheta_i) = -Q_{ji}, \quad \Lambda_{ij} = \Lambda_{ji} \geq 0,$$

and set $\pi_i = \sum_{j \neq i} Q_{ij}$. Then, for any network topology,

Two structural properties

$$\sum_{i \in \mathcal{D}} \pi_i = 0, \quad \sum_{i \in \mathcal{D}} \frac{\pi_i}{\vartheta_i} = \sum_{\{i, j\} \in \mathcal{E}} \Lambda_{ij} \frac{(\vartheta_j - \vartheta_i)^2}{\vartheta_i \vartheta_j} \geq 0.$$

RRHO assignments recover 2T and 4T models

Two temperatures

$$\mathfrak{t}\mathfrak{r} : Z_{\mathfrak{t}}^h Z_{\mathfrak{r}},$$

$$\mathfrak{v}\mathfrak{e} : Z_{\mathfrak{t}}^{e-} Z_{\mathfrak{v}} Z_{\mathfrak{e}}.$$

Four temperatures

$$\mathfrak{t} : Z_{\mathfrak{t}}^h,$$

$$\mathfrak{r} : Z_{\mathfrak{r}},$$

$$\mathfrak{v} : Z_{\mathfrak{v}},$$

$$\mathfrak{e} : Z_{\mathfrak{t}}^{e-} Z_{\mathfrak{e}}.$$

The modal assignment changes; the thermodynamic structure does not.

Park [6]; NIST RRHO treatment [11]

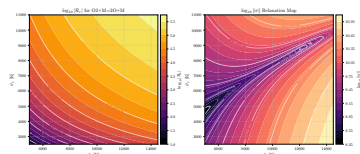
hyper-thermochem

hyper-thermochem is an energy-primary JAX-differentiable library that turns the gas model into thermodynamics, chemistry, transport, and source operators.

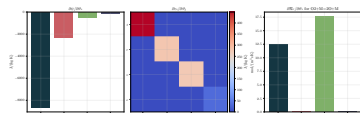
Differentiable Matrix-free sensitivities for implicit CFD solvers, obtained directly from the free energy.

Change the model Replace RRHO by RRMO—or another partition-function model—without changing the thermodynamic or CFD interface.

Scale Compile and batch the operators for parallel execution on CPUs or GPUs.



Chemistry over a two-temperature state space



AD sensitivities of chemistry and exchange

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Backup: Charge preserves the thermodynamic structure

Electrostatic microforce system

$$\begin{aligned}\operatorname{div} \boldsymbol{\xi}_\phi + \pi_\phi + \gamma_\phi &= 0, \\ \pi_\phi &= -\rho_e, \quad \boldsymbol{\xi}_\phi = -\epsilon \operatorname{grad} \phi, \\ \operatorname{div} (\epsilon \boldsymbol{E}) &= \rho_e.\end{aligned}$$

Electrochemical force and stress

$$\begin{aligned}\tilde{\mu}_{\alpha m} &= \mu_{\alpha m} + q_\alpha \phi, \\ \mathbf{T}^{\text{el}} &= \epsilon (\boldsymbol{E} \otimes \boldsymbol{E} - |\boldsymbol{E}|^2 \mathbf{1}/2), \\ \operatorname{div} \mathbf{T}^{\text{el}} &= \rho_e \boldsymbol{E}.\end{aligned}$$

The dissipation inequality keeps the neutral form with electrochemical potentials replacing chemical potentials.

Backup: two-temperature RRHO assignment

Two-temperature assignment

$$\mathfrak{m} = \mathfrak{tr}, \quad \mathcal{D} = \{\mathfrak{tr}, \mathfrak{ve}\},$$

$$\psi_{\mathfrak{m}}^{\text{mat}} = -\bar{R} \vartheta_{\mathfrak{tr}} \ln(Z_{\mathfrak{t}}^h Z_{\mathfrak{r}}),$$

$$\psi_{\mathfrak{ve}} = -\bar{R} \vartheta_{\mathfrak{ve}} \ln(Z_{\mathfrak{t}}^{e-} Z_{\mathfrak{v}} Z_{\mathfrak{e}}).$$

Free-electron translation is assigned to the vibration–electron mode.

Park [6]; NIST RRHO treatment [11]

Backup: four-temperature RRHO assignment

Four-temperature assignment

$$\mathbf{m} = \mathbf{t}, \quad \mathcal{D} = \{\mathbf{t}, \mathbf{r}, \mathbf{v}, \mathbf{e}\},$$

$$\begin{aligned} \psi_{\mathbf{m}}^{\text{mat}} &= -\bar{R} \vartheta_{\mathbf{t}} \ln Z_{\mathbf{t}}^h, & \psi_{\mathbf{r}} &= -\bar{R} \vartheta_{\mathbf{r}} \ln Z_{\mathbf{r}}, \\ \psi_{\mathbf{v}} &= -\bar{R} \vartheta_{\mathbf{v}} \ln Z_{\mathbf{v}}, & \psi_{\mathbf{e}} &= -\bar{R} \vartheta_{\mathbf{e}} \ln(Z_{\mathbf{t}}^e Z_{\mathbf{e}}). \end{aligned}$$

Free-electron translation is assigned to the electronic mode.

Park [6]; NIST RRHO treatment [11]

Backup: heat-flow conventions

Gurtin–Vargas variables

$$\mathbf{J}_q = \mathbf{q} + \sum_{\alpha \in \mathcal{C}} \mu_\alpha \mathbf{J}_\alpha, \quad \mathbf{J}_s = \frac{\mathbf{q}}{\vartheta}.$$

With partial specific entropy and enthalpy

$$\eta_\alpha = -\partial_{\vartheta} \mu_\alpha \big|_{p, \{\varphi_\beta\}}, \quad h_\alpha = \mu_\alpha + \vartheta \eta_\alpha,$$

de Groot–Mazur’s reduced heat flow is

$$\mathbf{J}'_q = \mathbf{J}_q - \sum_{\alpha \in \mathcal{C}} h_\alpha \mathbf{J}_\alpha = \mathbf{q} - \vartheta \sum_{\alpha \in \mathcal{C}} \eta_\alpha \mathbf{J}_\alpha.$$

Backup: both conventions give the same entropy flux

Substitution of the reduced heat flow gives

$$\mathbf{J}_s = \frac{\mathbf{J}'_q}{\vartheta} + \sum_{\alpha \in \mathcal{C}} \eta_\alpha \mathbf{J}_\alpha = \frac{\mathbf{q}}{\vartheta}.$$

The flux variables differ, but the entropy flux and entropy production are identical.

de Groot–Mazur [12]; Gurtin–Vargas [9]

Backup: Hansen–Fisher form

For an ideal-gas mixture,

$$\text{grad } \mu_\alpha = \frac{\bar{R}\vartheta}{\varphi_\alpha} \mathbf{d}_\alpha - \vartheta \eta_\alpha \text{grad } \ln \vartheta + \bar{R}\vartheta \text{grad } \ln p,$$

$$\mathbf{d}_\alpha = \text{grad } \xi_\alpha + (\xi_\alpha - \varphi_\alpha) \text{grad } \ln p.$$

The common pressure-gradient term is annihilated by the mobility null space, giving

$$\mathbf{j}_\alpha = -\bar{R}\vartheta \sum_{\beta \in \mathcal{C}} M_{\alpha\beta} \frac{\mathbf{d}_\beta}{\varphi_\beta} + \vartheta \sum_{\beta \in \mathcal{C}} M_{\alpha\beta} \eta_\beta \text{grad } \ln \vartheta.$$

Backup: diffusion, Soret and Dufour terms

$$\begin{pmatrix} \{\mathbf{J}_\alpha\}_c \\ \mathbf{J}'_q \end{pmatrix} = - \begin{pmatrix} M & -\vartheta M \boldsymbol{\eta} \\ -\vartheta \boldsymbol{\eta}^\top M & \kappa \vartheta + \vartheta^2 \boldsymbol{\eta} \cdot M \boldsymbol{\eta} \end{pmatrix} \begin{pmatrix} \{\bar{R} \vartheta \mathbf{d}_\alpha / \varphi_\alpha\}_c \\ \text{grad} \ln \vartheta \end{pmatrix}.$$

The temperature-gradient contribution to constituent diffusion is the Soret effect; its reciprocal contribution to the reduced heat flow is the Dufour effect.

Hansen–Fisher [7]